Umgesetzte Korrekturen: *One-parameter Semigroups of Positive Operators* (2. Aufl.)

Springer LNM 1184, Fahnenkorrektur zum Proof-PDF vom 14.10.2026. Stand: 14.10.2026.

Die Seitenangaben beziehen sich auf die gedruckten Buchseiten (PDF-Seite = Buchseite + 18).

Überblick

- Umgesetzt sind alle **122 Punkte mit hoher Sicherheit (●)** aus der Arbeitsliste `Fahnenkorrektur_Springer_LNM1184.md`, in den TeX- und Bib-Dateien unter `LN-Book-Springer`.
- Die Punkte mit mittlerer (◐) und niedriger (○) Sicherheit sind **noch offen**.
- **Merkposten A.4 („Part-I"):** Variante 1 umgesetzt. Die zwei Stellen auf S. 174 lauten jetzt „Part-I" wie im übrigen Fließtext.
- **Greiner/Voigt/Wolff (J. Operator Theory):** Der gemeinsame Bib-Eintrag hat jetzt Band 5. Damit sind auch die gleichlautenden ◐-Punkte auf S. 109, 132, 185, 367 und 463 erledigt.
- Formeln wurden nicht verändert.
- Die Änderungen liegen im Git-Branch `fahnenkorrektur`; `main` enthält weiterhin den Stand vor den Korrekturen.

Abweichungen von der Arbeitsliste

Bei diesen Punkten hat der Quelltext (Label-Namen) das gemeinte Verweisziel gezeigt; umgesetzt wurde daher etwas anderes als in der Spalte „Soll" der Arbeitsliste:

S.	Arbeitsliste	Umgesetzt	Begründung
107	Corollary 3.1.2 (?)	Part-III, Corollary 3.1.3	Label `cor:c3-1.3` verweist nach Teil III; „Part-III," fehlte.
114	Example 3.2.3/3.2.4 prüfen	Part-II, Example 3.1.7	Label `ex:b3-1.7` verweist nach Teil II; „Part-II," fehlte.
251	Part-I, Section 4.1.	Part-I, Section 4.1, Section 4.1 and Part-III, Section 4.1	Gemeint sind die Abschnitte 4.1 aus Teil I, II und III (Labels `sec:a4-1`, `sec:b4-1`, `sec:c4-1`).
257	Eq. (4.3.10)	the problem in Example 4.3.11	Das Label zeigt nur auf die Randbedingung; gemeint ist das ganze System des Beispiels.
461	Theorem 4.4.5 (iv)	Verweis auf Label `thm:d4-4.4`	Der Verweis geht jetzt auf den Satz, dessen Beweis an dieser Stelle steht.
468	Part-I, Section 3.6.6	the proof of Part-I, Theorem 3.6.6	Einen Abschnitt 3.6.6 gibt es nicht; im Beweis von Theorem 3.6.6 werden über das F-Produkt Näherungseigenvektoren zu Eigenvektoren gemacht.

Einzelkorrekturen

Teil I, Kap. 1 – Basic Results on Semigroups on Banach Spaces

S.	Ist	Soll	Hinweis
12	Let E to be one of the following function spaces	Let E be one of the following function spaces	
17	generator on a core of its domain (see Proposition 1.1.9 (ii). In particular	(see Proposition 1.1.9 (ii)). In particular	
23	form a strongly-continuos semigroup with respect to the sup-norm	form a strongly continuous semigroup with respect to the sup-norm	
26	In most cases F will be either the Frechét filter or some free ultra filter	In most cases F will be either the Fréchet filter or some free ultrafilter	Nur Fréchet/ultrafilter; die Klammerstellung (◐) ist noch offen.

Teil I, Kap. 2 – Characterization of Semigroups on Banach Spaces

S.	Ist	Soll	Hinweis
32	(which might not be known explicitely)	(which might not be known explicitly)	
36	So by Theorem 1.1.9 (ii) D(A) is a core of B.	So by Proposition 1.1.9 (ii) D(A) is a core of B.	
38	it follows that T(·)f is measureable.	it follows that T(·)f is measurable.	
45	(this follows from Theorem 1.1.6 (ii))	(this follows from Proposition 1.1.6 (ii))	
47	All together we obtain for \	r\	Doppelter Satz im Quelltext auskommentiert.
51	if and only if the following equivalent assertions holds.	if and only if the following equivalent assertions hold.	
58	Consequently it posseses a σ(E',E)-limit point	Consequently it possesses a σ(E',E)-limit point	
59	p(λR(λ,A)g) ≤ p(g)for all g ∈ E	p(λR(λ,A)g) ≤ p(g) for all g ∈ E	
60	(2t c‖g − Af‖ + t² c‖Ag‖) (by (2.3))	(by Eq. (2.2.2))	
70	[30] J. J. Moreau: Fonctionelles convexes. Séminaire sur les équations aux derivées partielles.	[30] J. J. Moreau: Fonctionnelles convexes. Séminaire sur les équations aux dérivées partielles.	

Teil I, Kap. 3 – Spectral Theory

S.	Ist	Soll	Hinweis
79	we recall the well known resolvent equation .	we recall the well-known resolvent equation.	
82	the spectral mapping theorem for the resolvent (Theorem 3.2.5)	(Proposition 3.2.5)	
100	a "multiplication operator"	a “multiplication operator”	
104	(cf. Formula Eq. (1.3.1) on page 19)	(cf. Eq. (1.3.1) on page 19)	
107	for some $k \in$ Z otherwise if ($\|R(\mu k, A)\|)k\in$Z is unbounded.	for some $k \in$ Z or if ($\|R(\mu k, A)\|)k\in$Z is unbounded.	
107	The Example 3.1.3 above in combination with Corollary 3.1.3 shows	Example 3.1.3 above in combination with Corollary 3.1.2 (?) shows	Umgesetzt als „Part-III, Corollary 3.1.3“ (siehe Abweichungen).
108	The spectral mapping theorem for the resolvent (Theorem 3.2.5) is well known	(Proposition 3.2.5)	
108	can be found in Kato [16], Chap.III,§ 6.4].	can be found in Kato [16, Chap. III, § 6.4].	
108	Applications to isolated singularities like Example 3.6 are discussed	like Example 3.3.6 are discussed	

Teil I, Kap. 4 – Asymptotics of Semigroups on Banach Spaces

S.	Ist	Soll	Hinweis
114	for all $f \in C$0(Rn); see also Example 3.1.7.	Verweisziel prüfen (z. B. Example 3.2.3/3.2.4)	Umgesetzt als „Part-II, Example 3.1.7“ (siehe Abweichungen).
119	Example 4.1.2(c) shows that uniform exponential stability is not equivalent to	Example 4.1.2 (ii) shows	
121	On the other hand, Example 4.1.2(c) produces a semigroup	Example 4.1.2 (ii) produces a semigroup	
125	The Laplacian Δ generates a bounded holomorphic semigroups on Lp(Rn)	generates a bounded holomorphic semigroup on	

Teil II, Kap. 1 – Basic Results on Spaces $C_0(X)$

S.	Ist	Soll	Hinweis
139	This identification is a topological isomorphisms with respect to the weak*-topology	This identification is a topological isomorphism with respect to the weak* topology	
139	(i) positive (notation: $T \geqslant 0$) if $T f$ is a positive whenever f is positive,	if Tf is positive whenever f is positive,	

Teil II, Kap. 2 – Characterization of Positive Semigroups on $C_0(X)$

S.	Ist	Soll	Hinweis
143	It will be essential that K is compact for all what follows	It will be essential that K is compact for all that follows	
148	The next result is a strengthened form of Theorem 1.6.	The next result is a strengthened form of Theorem 2.1.6.	
152	preceding page succesively to B = m^{1/k} · A	preceding page successively to …	
154	Then A is the generator of positive semigroup.	Then A is the generator of a positive semigroup.	
161	hence T(t) is so as well (cf. Remark 2.1).	hence T(t) is so as well (cf. Remark 2.2.1).	
161	we denote by dϕ he linear form on C[−1, 0]	we denote by dϕ the linear form on C[−1, 0]	
163	this will enable us, to give a detailed description	this will enable us to give a detailed description	
167	is closable (by Corollary 2.1.11); but	is closable (by Corollary 2.1.11); but	
179	is closed in (a, b) Thus (a, b) ∖ K is the union	is closed in (a, b). Thus (a, b) ∖ K is the union	
183	It is easy to verify, that the generator of this group is	It is easy to verify that the generator of this group is	
184	we refer to the corresponding notes in Part-III, Chapter 2, Theorem 2.1.6 and 2.1.8 are due to Arendt et al. [4]	… in Part-III, Chapter 2. Theorems 2.1.6 and 2.1.8 are due to Arendt et al. [4]	
184	Infinite-dimensional Systems, Retahof 1983.	Infinite-dimensional Systems, Retzhof 1983.	
185	[11] P. Collatz: Einschließungssatz für die charakteristischen Zahlen von Matrizen.	[11] L. Collatz: Einschließungssatz …	

Teil II, Kap. 3 – Spectral Theory of Positive Semigroups on $C_0(X)$

S.	Ist	Soll	Hinweis
202	A real spectral value λ0 of a generator A is called a dominant provided that	… is called dominant provided that	
204	unless ϕ = r0δ0 for some r0 ∈ R.)	unless ϕ = r0δ0 for some r0 ∈ R.	
212	such that Pσ(A) = H). Here (H_d)^ denotes	such that Pσ(A) = H. Here (H_d)^ denotes	
215	A complete proof of this results requires	A complete proof of this result requires	
220	by Part-I, Proposition 3.6.2 Since every boundary point	by Part-I, Proposition 3.6.2. Since every boundary point	
221	by Lemma 3.4.2 (b) there exists y ∈ K∞	by Lemma 3.4.2 (ii) there exists y ∈ K∞	
221	For details see Sect.18.2 of Dieudonné [11].	For details see Section 18.2 of Dieudonné [11].	
222	(iii) We consider Ω as in (b) and assume	(iii) We consider Ω as in (ii) and assume	
222	by remarking that the ideas of (b) and (c) can be combined	by remarking that the ideas of (ii) and (iii) can be combined	
227	The dichotomy stated in (Eq.(3.2.19) on page 202) is probably	The dichotomy stated in Eq. (3.2.19) on page 202 is probably	
229	[22] J. P. Roth: “Les operateurs elliptiques comme générateurs infinitésimaux de semigroups de Feller”. Séminaire de Théorie du Potential	“Les opérateurs elliptiques comme générateurs infinitésimaux de semi-groupes de Feller”. Séminaire de Théorie du Potentiel	

Teil II, Kap. 4 – Asymptotics of Positive Semigroups on $C_0(X)$

S.	Ist	Soll	Hinweis
240	is a countable additive set function	is a countably additive set function	
242	Then ω_ess(T) < 0 by Eq. (3.3.10) on page 85).	Then ω_ess(T) < 0 by Eq. (3.3.10) on page 85.	
251	as developed in Part-I, Section 4.1, Section 4.1	as developed in Part-I, Section 4.1.	Umgesetzt als „Part-I, Section 4.1, Section 4.1 and Part-III, Section 4.1“ (siehe Abweichungen).
253	then σ(A) = ∅ by Proposition 3.4.	then σ(A) = ∅ by Proposition 4.3.4.	
257	Then (3.11) takes the abstract form	Then Eq. (4.3.10) takes the abstract form	Umgesetzt als „Then the problem in Example 4.3.11 takes …“ (siehe Abweichungen).
259	showed the translation property (T)for the solution semigroup	showed the translation property (T) for the solution semigroup	
259	(see Part-III, Section 3.4 can be used to derive	(see Part-III, Section 3.4) can be used to derive	

Teil III, Kap. 1 – Basic Results on Banach Lattices and Positive Operators

S.	Ist	Soll	Hinweis
265	Rainer Nagel und Ulf Schlotterbeck	Rainer Nagel and Ulf Schlotterbeck	
267	contains x1, ..., xn (see Section. 1.4).	contains x1, ..., xn (see Section 1.4).	

Teil III, Kap. 2 – Characterization of Positive Semigroups on Banach Lattices

S.	Ist	Soll	Hinweis
284	of all infinitely differentiabele functions on Ω	of all infinitely differentiable functions on Ω	
292	It has been proved by Miyajima & Okazawa [23]] that	It has been proved by Miyajima & Okazawa [23] that	
294	In the proof of the "only if" - part of Proposition 2.3.6	In the proof of the “only if” part of Proposition 2.3.6	
303	which finally implies that B is closable Now let λ > λ0	which finally implies that B is closable.	
315	is right-sided Gateaux differerentiable and	is right-sided Gateaux differentiable and	
325	[27] J. Peetre: Une charactérisation abstraite des opérateurs différentiels.	[27] J. Peetre: Une caractérisation abstraite des opérateurs différentiels.	

Teil III, Kap. 3 – Spectral Theory of Positive Semigroups on Banach Lattices

S.	Ist	Soll	Hinweis
341	then T(t)f is a quasi-interior pointof E+ whenever	then T(t)f is a quasi-interior point of E+ whenever	
348	which is finer than the Frechet filter	which is finer than the Fréchet filter	
350	ultrafilter U on N which is finer than the Frechet filter	ultrafilter U on N which is finer than the Fréchet filter	
352	by calculating the resolvent explicitely	by calculating the resolvent explicitly	
354	appearing in the the representation Eq. (3.3.21)	appearing in the representation Eq. (3.3.21)	
357	the Césaro means	the Cesàro means	
366	or “quasi-strictly positive” in Karlin [21]). Sawashima [30]	or “quasi-strictly positive” in Karlin [21]. Sawashima [30]	
368	[34] J. Voigt: On the abszissa of convergence for the Laplace transform	On the abscissa of convergence for the Laplace transform	

Teil III, Kap. 4 – Asymptotics of Positive Semigroups on Banach Lattices

S.	Ist	Soll	Hinweis
370	$s(A) \in \sigma(A)$ (see Corollary 3.1.4).	(see Corollary 3.1.4).	
385	a Markov transition function satisfying all the assumptions of 4.2.7 induces	a Markov transition function satisfying all the assumptions of Example 4.2.8 induces	
402	(see Schaefer [36, Section II.5] and thus compactness of $B2$	(see Schaefer [36, Section II.5]) and thus compactness of $B2$	
402	As in Example 4.3.13 the first equation in (4.3.14) corresponds to the aging process.	As in Example 4.3.12 the first equation in (4.3.14) corresponds to the aging process.	

Teil IV, Kap. 1 – Basic Results on Semigroups and Operator Algebras

S.	Ist	Soll	Hinweis
413	Rp has both the Schwarz type property and is identity preservation.	Rp has the Schwarz type property and is identity preserving.	
414	the following equalities holds	the following equalities hold	

Teil IV, Kap. 2 – Characterization of Positive Semigroups on W*-Algebras

S.	Ist	Soll	Hinweis
418	Using the Schwarz, inequality we obtain	Using the Schwarz inequality, we obtain	

Teil IV, Kap. 3 – Spectral Theory on W*-Algebras and their Preduals

S.	Ist	Soll	Hinweis
422	subinvariant for a positive map $T \in \mathrm{L}(M)$ (resp.. positive semigroup T)	(resp. positive semigroup T)	
425	by the uniqueness theorem (*) on p. 422 for the absolut value	for the absolute value	
426	we obtain that φ+ and φ− in Fix(R).	we obtain that φ+ and φ− are in Fix(R).	
432	3.2 Spectral Properties of Uniformly Ergodic Semigoups	3.2 Spectral Properties of Uniformly Ergodic Semigroups	
432	the discret group $P\sigma(A) \cap$ iR	the discrete group	
434	and the assertion follows from Lemma 3.1.6).	and the assertion follows from Lemma 3.1.6.	
437	and the assocciated ergodic projection	and the associated ergodic projection	
438	Our main result Theorem 3.1.10 on extends Part-II, Theorem 3.3.6	Our main result Theorem 3.1.10 extends Part-II, Theorem 3.3.6	
439	G. Greiner, J. Voigt & M. Wolff: On the spectral bound … J. Operator Theory (1981) 245–256.	J. Operator Theory 5 (1981) 245–256.	Band 5 im gemeinsamen Bib-Eintrag ergänzt; wirkt in allen Kapiteln.

Teil IV, Kap. 4 – Asymptotics on C- *and W*-Algebras

S.	Ist	Soll	Hinweis
442	characterizes uniform exponentially stability of the semigroup	characterizes uniform exponential stability	
443	Then the following assertion are equivalent.	Then the following assertions are equivalent.	
445	whereas (c) and (d)), resp. (e) and (f) are equivalent by the Remark 4.2.1 (iii).	whereas (c) and (d), resp. (e) and (f), are equivalent by Remark 4.2.1 (iii).	
446	(see Propositions 4.3.7, 4.3.8 &. 4.3.9)	(see Propositions 4.3.7, 4.3.8 & 4.3.9)	Umgesetzt als „Proposition 4.3.7, Theorem 4.3.8 and Corollary 4.3.9“.
446	Therefore the following compactness criterium is quite useful.	compactness criterion	
446	the orbit $\{T(t)\varphi : t \in \mathrm{R}+\}$ is relatively weak compact	is relatively weakly compact	

448	(see Krengel [14, Chap.2, Theorem1.4])	(see Krengel [14, Chap. 2, Theorem 1.4])	
449	By Proposition 3.1 the semigroup T is strongly ergodic.	By Proposition 4.3.1 the semigroup T is strongly ergodic.	
451	If $u1$ and $u2$ in U, then	If $u1$ and $u2$ are in U, then	
451	the fact that $T0(t)$' is a *-automorphismus on N.	is a *-automorphism on N.	
452	as group of *-automorphism such that the fixed space	as a group of *-automorphisms	
452	We apply now these results to weak*-semigroup on B(H).	We now apply these results to weak*-semigroups on B(H).	
452	and φ is a faithful T-invariant normal states.	a faithful T-invariant normal state.	
452	irreducible (equvalently, if the fixed space	(equivalently, if	
454	a predual M_* of a von Neumann-algebra M	of a von Neumann algebra M	
459	Without further comments, we will now use following facts.	we will now use the following facts.	
461	hence Theorem 4.4.2 (iv) is proved.	hence Theorem 4.4.5 (iv) is proved.	Verweis auf den Satz, dessen Beweis hier steht (siehe Abweichungen).
461	exists in the strong operator topology with A' is the generator of T'.	where A' is the generator of T'.	
463	U. Groh & B. Kümmerer: Bibounded operators W_*-algebras.	Bibounded operators on W$_*$-algebras.	

Updated Notes

S.	Ist	Soll	Hinweis
468	to the spectral theory of $C0$-semigroups, as in Chapter 3.6.6, was started	as in Part-I, Section 3.6.6	Umgesetzt als „the proof of Part-I, Theorem 3.6.6“ (siehe Abweichungen).
468	Using this, El-Mennaoui & Engel [30] gives a characterization	El-Mennaoui & Engel [30] give a characterization	
470	Quantative estimates of the decay of the semigroup	Quantitative estimates	
471	[16] J. Diestel: A survey of results related to the Dunford-Pettis property. ... Contemp. Math. 2, 15-60 (1980). 1980.	... Contemp. Math. 2 (1980) 15–60.	Bib-Eintrag auf @InCollection umgestellt.
474	As examples for the rich litterature we mention	the rich literature	
475	persistent irreducibilty implies that the generator	persistent irreducibility	
475	In Gerlach [21] strong convergence of Feller semigoups is studied.	Feller semigroups	
481	Even the spectrum of the generator is independent on p in this case	independent of p	
481	The domain of its generator can be described explicitely	explicitly	
489	This also follows from the example given above above and the fact	from the example given above and the fact	
490	By Greiner et al. [12, Theorem 5.3.1]] one has	By Greiner et al. [12, Theorem 5.3.1] one has	